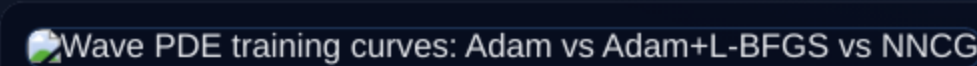


Training PINNs

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Adam converges slowly; Adam+L-BFGS stalls ~40k steps; NNCG after Adam+L-BFGS further improves.

ABSTRACT

- PINN loss is ill-conditioned due to differential operators.
- Adam+L-BFGS beats Adam or L-BFGS; NNCG further improves.
- Theory links operator and loss conditioning; supports hybrids.

OPTIMIZER

Adam+L-BFGS best

NEW METHOD

NNCG

EFFECT

Lower loss & L2RE

INTRODUCTION

- PINNs solve PDEs via residual, boundary, and initial losses.
- Mesh-free promise vs hard optimization from ill-conditioning.
- We study the loss landscape and improve training with hybrids.

KEY FINDING

- Single sentence: Even after Adam+L-BFGS stalls, a brief NNCG phase slashes loss and L2RE by another order of magnitude.
- Explanation: L-BFGS halts under strong Wolfe before a critical point ($\text{grad} \approx 1\text{e-}2\text{--}1\text{e-}3$); NNCG's damped Newton steps with Armijo and Hessian-vector products exploit remaining curvature to make decisive progress.

CONTRIBUTIONS

- Hessian spectra reveal severe ill-conditioning; quasi-Newton improves $\geq 1000\times$.
- Adam+L-BFGS outperforms Adam or L-BFGS across problems.
- Accurate solutions demand near-zero loss; NNCG boosts further.
- Theory: operator ill-conditioning $\Rightarrow$ loss ill-conditioning; hybrids justified.

SETUP

- Model $u(x; w)$ with tanh MLP; minimize residual and BC/IC losses.
- Interpolation (loss ≈ 0) is feasible and desirable for generalization.
- Common pipeline: Adam then L-BFGS.

 Training pipeline: sampling $\rightarrow$ model $\rightarrow$ losses $\rightarrow$ optimizer switches

Pipeline schematic: dataset sampling $\rightarrow$ PINN (MLP) $\rightarrow$ residual/BC/IC losses $\rightarrow$ Adam $\rightarrow$ L-BFGS ($\rightarrow$ NNCG).

METHODS

- Convection, wave PDEs; reaction ODE (challenging settings).
- MLPs: 3 layers; widths 50/100/200/400; Xavier init.
- Sampling: 10k residual; dense IC/BC; metric: L2RE.
- Optimizers: tuned Adam; L-BFGS (lr=1.0, m=100); switches at 1k/11k/31k; total 41k iters.
- Env: PyTorch 2.0, Python 3.10, Titan V, CUDA 11.8.

LOSS VS ERROR

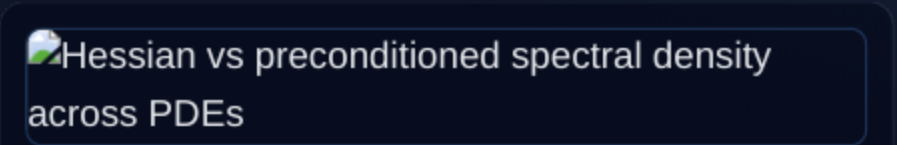
- Lower training loss $\rightarrow$ lower L2RE across PDEs.
- Convection: loss $1e-3 \rightarrow$ L2RE $\approx 1e-1$; $1e-5 \rightarrow \approx 1e-2$.

 Scatter: final L2RE vs final loss across runs

Lower loss correlates with better accuracy (Figure 2).

ILL-CONDITIONING

- Hessian spectra: large outliers and mass near 0.
- Residual term (with D) dominates ill-conditioning.
- L-BFGS preconditioning cuts top eigenvalues $\geq 1e3$.



Hessian vs preconditioned spectral density across PDEs

Convection/Reaction/Wave: severe ill-conditioning; quasi-Newton improves (Figure 3 top).



Component-wise spectra: residual, IC, BC

Residual dominates; all components improved by L-BFGS (Figure 3 bottom).

OPTIMIZERS

- Adam+L-BFGS achieves lowest loss and L2RE across widths and PDEs.
- Examples (Table 1): Convection L2RE $5.96e-2$ (Adam) $\rightarrow$ $4.19e-3$ (Adam+L-BFGS); Wave $3.35e-1$ (L-BFGS) $\rightarrow$ $5.52e-2$.

ACROSS PDES

Combined wins

REASON

Avoid saddles + faster local rate

EXCEPTION

Reaction milder

HYBRID + NNCG

- First-order avoids saddles but slows when κ is large; quasi-/second-order speed locally.
- Adam $\rightarrow$ L-BFGS: escape saddles, then reduce κ ; if stalled, switch to NNCG.
- NNCG: damped Newton with Armijo; Hessian-free via Hvps; Nyström-PCG preconditioning.



Diagram: Adam $\rightarrow$ L-BFGS $\rightarrow$ NNCG with line searches and preconditioning

Pipeline with NNCG fine-tuning for high precision.

UNDER-OPTIMIZED

- L-BFGS with strong Wolfe may pick step size 0 and stop before a critical point.

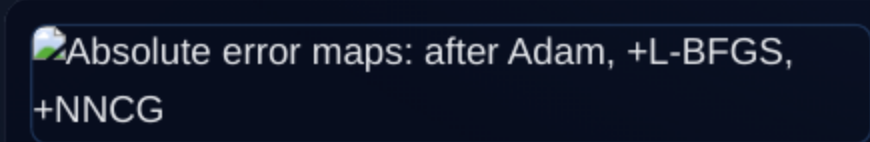


Training curves showing L-BFGS early stop and nonzero gradients

Runs stall with grad norms $\approx 1e-2$ – $1e-3$ (Figure 4/9).

NNCG RESULTS

- After Adam+L-BFGS, NNCG reduces loss $>10\times$ and lowers gradients; GD gives no gains.



Absolute error maps: after Adam, +L-BFGS, +NNCG

Progressive error reduction (Figure 5).

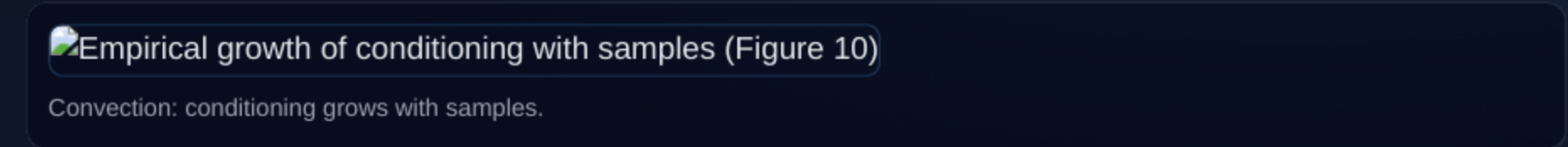


Bar chart of Loss/L2RE: Adam+L-BFGS vs +NNCG vs +GD

NNCG best; GD $\approx$ no change (Table 2).

OPERATOR $\rightarrow$ LOSS

- Ill-conditioned D induces ill-conditioned PINN loss.
- Theorem 8.4 (informal): spectrum of $A \cdot K_\infty(w^\star)$ controls conditioning; polynomial decay $\Rightarrow \kappa$ grows with n_{res} .

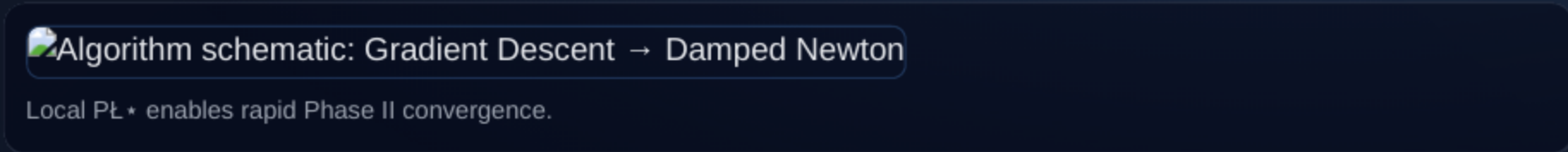


Empirical growth of conditioning with samples (Figure 10)

Convection: conditioning grows with samples.

GDND RATE

- GDND: switch from GD to damped Newton; Phase II has linear rate independent of κ (Theorem 8.5).



Algorithm schematic: Gradient Descent $\rightarrow$ Damped Newton

Local PL $\star$ enables rapid Phase II convergence.